

# Software Engineering with Generative AI

Michigan Technological University — Fall 2026 CS 5090 (3 credits, graduate)

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|---------------------|------------------------------------------------------------------------------|
| Instructor          | Dr. Jie J.W. Wu                                                              |
| Email               | jie.jw.wu@mtu.edu                                                            |
| Office hours        | Mon/Wed 9:30–10:30am, Rekhi 207 — please make an appointment before visiting |
| Meeting time / room | Mon/Wed/Fri 12:00 – 12:50pm, Electrical Energy Resources, Room 0227          |
| Semester            | Aug 31 – Dec 11, 2026 (no final exam)                                        |

## MTU Fall 2026 Key Dates

Per the official [MTU 2026–27 academic calendar](#):

| Date               | Event                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mon, Aug 31        | First day of classes (Week 1)                                                                                                                                                |
| Fri, Sep 18        | Project 2 topics circulated — the research thread starts in Week 3                                                                                                           |
| Fri, Oct 2         | <b>Project 2 research radar due</b> — solo-or-pair declared, two candidate directions, three papers skimmed                                                                  |
| Wed, Oct 7         | <b>Midterm 1</b>                                                                                                                                                             |
| Fri, Oct 23        | Pooled-findings session in class (bring your current Project 1 findings table)                                                                                               |
| Mon, Oct 26        | Project 2 topic pitches — class held <b>online</b>                                                                                                                           |
| Wed, Oct 28        | Class held <b>online</b>                                                                                                                                                     |
| Fri, Oct 30        | <b>No class — deadline day. Project 1 due in full</b> (one deadline: all four phases + demo video) • <b>Project 2 proposals due</b> — topic pivots allowed until Fri, Nov 20 |
| Wed, Nov 18        | <b>Midterm 2</b> (covers Module 2) — the latest slot MTU exam policy allows a clear run after                                                                                |
| Fri, Dec 4         | Project 2 full paper drafts due                                                                                                                                              |
| Mon & Wed, Dec 7–9 | <b>Project 2 design presentations + preliminary results</b> (in class)                                                                                                       |
| Fri, Dec 11        | Last day of classes; <b>Project 2 final paper due</b>                                                                                                                        |
| Dec 14–18          | Finals week — <b>no final exam in this course</b>                                                                                                                            |

Holiday class cancellations are noted in the Course Schedule below. Exact midterm days will be adjusted to the actual meeting pattern once scheduled; add/drop and withdrawal deadlines follow the Registrar's calendar.

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## Course Description

Generative AI — and large language models (LLMs) in particular — is materially changing how software is built. This course examines that change from both directions: **software engineering of generative AI systems** (building, evaluating, and operating systems with learned components) and **software engineering with generative AI** (AI-assisted and agentic development workflows).

AI-assisted and agentic software engineering is not a standalone unit at the end — it is a **thread that runs through the whole course**. From Week 1, students use AI coding assistants and agents as their required development workflow (and learn to validate their output); the foundations module explains how those tools work by building one; the production module covers how to engineer, evaluate, and operate LLM-powered and agentic systems. The second half of the course runs as a research seminar: students — solo or in pairs — carry out an original research project and write it up as a conference-format paper, which is submitted to a suitable venue after the semester ends.

### Prerequisites

- Probability and statistics, linear algebra, data structures
- Programming maturity (Python); prior exposure to machine learning is helpful but not required
- Graduate standing, **or instructor permission for undergraduates**

### Undergraduate students

This is a graduate course, and a small number of undergraduates take it with instructor permission. They do the same lectures, the same Project 1, and the same exams. Three adjustments are **recommended — suggestions, not requirements** — so that the graduate-level research expectations land more gently on a student who has not been taught them yet:

| Recommended adjustment     |                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project 2 pairing</b>   | Project 2 is done solo or in pairs; undergraduates are <b>encouraged to pair with a graduate student</b> rather than working solo. |
| <b>Design presentation</b> | Undergraduates paired with a graduate student may <b>co-present</b> the design presentation rather than presenting alone.          |
| <b>Reading load</b>        | One paper per week is a reasonable target where graduate students read two; the second is optional.                                |

Everything else — Project 1 in full, both midterms, the findings table, the Week 8 pooled-findings session — is identical. Project 1 is deliberately scaffolded phase by phase and does not assume research experience, so it is the right entry point regardless of level.

The workshop-submission bonus is open to undergraduates on the same terms as everyone else.

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## Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Explain at a principled level how large language models are trained (pretraining, instruction tuning, RLHF/preference optimization) and how they generate output.
2. Build, fine-tune, and evaluate a small code-generation language model end to end, including data preparation, tokenization, training, and benchmark evaluation.
3. Analyze training datasets for provenance, bias, contamination, licensing, and governance implications.
4. Explain embeddings as a representational framework and apply them to retrieval and similarity tasks (e.g., retrieval-augmented generation over codebases).
5. Evaluate AI as a system component with defined cost, latency, reliability, safety, and privacy characteristics, and design systems that mitigate common failure modes (hallucination, adversarial inputs, miscalibration, prompt injection).
6. Apply production ML practices — requirements, architecture, deployment, testing, monitoring, and MLOps — to systems with LLM components.
7. Effectively use AI coding assistants and AI agents within a disciplined, collaborative development workflow (version control, code review, validation of AI-generated code), and design and evaluate agentic systems for SE tasks.
8. Critically read and discuss current research in AI-assisted and agentic software engineering.
9. Design and execute an empirical software engineering study involving generative AI, and communicate the results in a conference-quality research paper.

## Course Materials

**Two textbooks, used deeply — and no purchase is needed.** The free materials listed below fully cover this course. T1 is a Manning book, but its companion repository (the complete, runnable code for every chapter) and the resources under it are enough on their own. **If you are able and want to support T1's author — Sebastian Raschka, a former UW–Madison professor — buying the book is a fine way to do it.** T2 is freely readable online, and every other reading is a research paper available via arXiv or the ACM/IEEE digital libraries (through the MTU library).

### Textbooks

| #  | Text                                                                                                                                                                     | Used for                                                                                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| T1 | S. Raschka, <i>Build a Large Language Model (From Scratch)</i> , Manning, 2024                                                                                           | Module 1 and Project 1: tokenization, attention, the GPT architecture, pretraining, fine-tuning — the same build-it path the course takes |
| T2 | C. Kästner, <i>Machine Learning in Production: From Models to Products</i> , MIT Press — <a href="https://mlip-cmu.github.io/book/">https://mlip-cmu.github.io/book/</a> | Module 2: requirements, architecture, quality assurance, operations, responsible AI                                                       |

**T1 study materials** — free, and fully sufficient for this course:

- Official companion code, chapter by chapter: <https://github.com/rasbt/LLMs-from-scratch>
- The author's page for the book: <https://sebastianraschka.com/llms-from-scratch/>

- Community study notes: [https://knowledge.zhaoweiguo.com/build/html/x-learning/books/ais/2024/build\\_llm\\_from\\_scratch](https://knowledge.zhaoweiguo.com/build/html/x-learning/books/ais/2024/build_llm_from_scratch)
- A community re-implementation, notebook by notebook: <https://github.com/codewithdark-git/Building-LLMs-from-scratch>

### Additional books — free, optional

- T. Xiao & J. Zhu, *Foundations of Large Language Models* (2025) — <https://arxiv.org/pdf/2501.09223> — free; pre-training, generative models, prompting and alignment in one place. The closest companion to Module 1's second half
- A. E. Hassan et al., *Agentic Software Engineering: A Practical Guide for the AI-Native Engineer* — <https://book.agentic-swe.dev/> — the closest companion to the ⚙ thread
- A. Zhang, Z. C. Lipton, M. Li, A. J. Smola, *Dive into Deep Learning* — <https://d2l.ai/> — deeper neural-network foundations, with runnable code

The ⚙ AI-assisted / agentic SE thread is grounded directly in the research literature (e.g., Hassan et al.'s agentic-SE roadmap below, and their book above) rather than an assigned textbook — the field moves faster than books do.

### Research papers

A **suggested reading list** — mostly software engineering research, with the AI papers needed to make sense of it, and a few of the instructor's own — is published on the course website: [Reading list](#). It is a menu, not a required set: each week's readings pair a textbook chapter with 1–2 papers drawn from it, and every slide making a specific claim shows its source at the foot of the slide.

## Course Schedule

**Two teaching modules with the ⚙ AI-assisted/agentic SE thread woven through both, then a flexible design-review block. Three 50-minute lectures per week (Mon/Wed/Fri)** when there is no holiday. ×2 means a topic spans two sessions; ⚙ marks a practice session from the AI-assisted/agentic SE thread — supporting the projects, never examined. Slide PDFs are posted on the course website after each class.

**Sessions are subject to change.** The week-by-week content below is the plan, not a contract: pacing is adjusted to how the class is actually going, and current events in a fast-moving field sometimes earn a session. Dated deadlines and exam dates change only if announced in class *and* on the course website.

**Holiday cancellations (MTU academic calendar):** no class **Mon, Sep 7** (Labor Day) — Week 2 has **two sessions** — and no classes the week of **Nov 23–27** (Thanksgiving). Marked 🚫 below.

### Module 1 — Foundations: Build a Code LM, with the ⚙ thread (Weeks 1–8)

Ten foundations lectures plus three ⚙ practice sessions, each ⚙ session landing just before the Project 1 phase it supports. Slides are posted on the course website after each class.

T1 (Raschka) is read in course order and **finished by Week 5, on purpose** — from Week 7 the readings are research papers, because that is where the field lives. The **one idea** column is what to keep if you keep nothing else.

| Week | Dates          | Sessions                                                                                                                                                                                   | The one idea                                                             | Read                                                                          | Projects                                                          |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1    | Aug 31 – Sep 4 | Course overview · ☸ <b>hands-on: assistant, Git, compute, AI-use log</b> · what a language model is                                                                                        | Tokens, embeddings, next-token prediction                                | T1 ch. 1–2                                                                    | <b>Phase 1 released</b>                                           |
| 2    | Sep 8 – 11     | Attention and the Transformer block · position, norms, model families                                                                                                                      | <b>Attention and the Transformer block</b> — the core idea of the course | T1 ch. 3–4 · Vaswani et al. 2017                                              | 🚫 Labor Day — two sessions                                        |
| 3    | Sep 14 – 18    | Decoding, guided output, KV cache, MoE, prompting ×2 · ☸ <b>building with AI assistance</b> — context engineering, decomposition, plan-then-act                                            | Decoding and prompting — the model as an interface you tune at run time  | T1 ch. 5 · Chen et al. 2021 (Codex)                                           | training runs start · <b>P2 topics circulated Fri Sep 18</b>      |
| 4    | Sep 21 – 25    | Pretraining, data mix, scaling laws, SFT and LoRA ×2 · Project 1 clinic                                                                                                                    | <b>Pretraining and scaling</b> — where capability comes from; LoRA       | T1 ch. 5–6 + App. E · Hoffmann et al. 2022 · Hu et al. 2021                   | Build pace: Sep 25 (recommended)                                  |
| 5    | Sep 28 – Oct 2 | Preferences, reward models, DPO · reasoning, test-time scaling, GRPO · ☸ <b>the new workflow</b> — author to reviewer, levels of autonomy, the evidence                                    | Instruction following, preferences, and reasoning models                 | T1 ch. 7 · Ouyang et al. 2022 · Rafailov et al. 2023 · DeepSeek-R1            | <b>P2 radar due Fri Oct 2</b> — solo/pair + directions + 3 papers |
| 6    | Oct 5 – 9      | Midterm review · <b>Midterm 1 (Wed, Oct 7)</b> · Phase 2 debrief                                                                                                                           | Revise from each lecture's Key-takeaways slide                           | —                                                                             | Use pace: Oct 9 (recommended)                                     |
| 7    | Oct 12 – 16    | Retrieval over a repo, tool calling, MCP, the agent loop ×2 · ☸ <b>agentic SE</b> — a worked episode, tool design, SWE-bench, failure modes                                                | <b>Beyond the textbook</b> — retrieval, tools, the agent loop            | Lewis et al. 2020 · Yao et al. 2023 (ReAct) · Jimenez et al. 2024 (SWE-bench) | Break pace: Oct 16 (recommended)                                  |
| 8    | Oct 19 – 23    | Evaluation: pass@k, benchmarks, contamination · ☸ <b>testing with LLMs</b> — oracle problem, TDD, mutation score · <b>pooled-findings session</b> (bring your current table) + module wrap | Evaluating code models honestly                                          | Chen et al. 2021 §2.1 · one leaderboard's contamination statement             | Assure pace: Oct 23 (recommended)                                 |

**Midterm 1 covers the Weeks 1–6 lectures** (through reasoning); the Weeks 7–8 material and the ☸ sessions are not examined on it.

## Module 2 — Machine Learning in Production, with the ☸ thread (Weeks 9–11)

Eight production lectures with three ☸ pairings — each ☸ session taught alongside the production topic it extends, selecting from both. Slides are posted on the course website after each class.

T2 (Kästner) is read by chapter cluster, matching the lectures; each ☸ pairing adds one paper from the slide citations.

| Week | Dates       | Sessions                                                                                                                                                                                      | The one idea                                                                   | Read (T2 chapters)                                                                                     | Projects                                                        |
|------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 9    | Oct 26 – 30 | <i>Travel week — Mon and Wed held online. P2 pitches (Mon, online) · the model is a small box (Wed, online) · 🚫</i><br><b>Fri: no class — deadline day</b>                                    | <b>The model is a component</b> — systems thinking around a learned part       | From Models to Systems                                                                                 | <b>P1 due in full + P2 proposal due Fri Oct 30</b>              |
| 10   | Nov 2 – 6   | When ML is the right tool, goals and measurement · requirements and mistakes + 🌀 <b>requirements in the AI era</b> (paired, select) · architecture, deployment, MLOps                         | Goals, requirements, <b>planning for the model being wrong</b> ; architecture  | when-to-use-ML and goal-setting · requirements + planning-for-mistakes · architecture/deployment/MLOps | <b>experiments start Mon Nov 2</b> — proposal feedback returned |
| 11   | Nov 9 – 13  | Four-level QA + 🌀 <b>assuring generated code</b> (paired, select) · testing in production, versioning · process, teams, debt + 🌀 <b>productivity</b> and responsible engineering (selections) | <b>Quality assurance at four levels</b> ; production; teams and responsibility | model-quality + data-quality · testing-in-production · process/teams/debt + responsible-engineering    | experiments running · team check-ins in office hours            |

**Midterm 2 covers the Module 2 lectures** (taught through Fri Nov 13). The 🌀 pairings are not examined.

### Weeks 12–14 — Midterm 2, advanced topics, then the Project 2 finish

The end of the course runs: **exam** → **advanced SE topics** → **draft** → **design presentation with preliminary results** → **final paper**. The advanced-topic sessions are flexible — they are taught from the 🌀 deep-dive material and current papers, and any of them yields to a Project 2 clinic when teams need it.

| Session       | Plan                                                                                                          |
|---------------|---------------------------------------------------------------------------------------------------------------|
| Mon, Nov 16   | Midterm 2 review session                                                                                      |
| Wed, Nov 18   | <b>Midterm 2</b> — covers the Module 2 decks, taught through Fri Nov 13                                       |
| Fri, Nov 20   | 🌀 <b>Advanced topic 1</b> — agentic SE, the full deep dive                                                    |
| 🚫 Nov 23 – 27 | <i>Thanksgiving — no classes</i>                                                                              |
| Mon, Nov 30   | 🌀 <b>Advanced topic 2</b> — AI for testing and assurance, in depth                                            |
| Wed, Dec 2    | 🌀 <b>Advanced topic 3</b> — instructor's pick from the current literature, or by class vote                   |
| Fri, Dec 4    | Paper clinic · <b>draft due (end of day)</b>                                                                  |
| Mon, Dec 7    | <b>Design presentations + preliminary results</b> , first block (8 min + 2 questions; five talks per session) |
| Wed, Dec 9    | <b>Design presentations + preliminary results</b> , second block                                              |
| Fri, Dec 11   | Course wrap — the professional stance · <b>final paper + artifact due</b>                                     |

**Why the advanced topics sit there.** They are cutting-edge material — agents, AI-for-testing, whatever the literature is doing this month — taught when exams are over and students are deep in their own research. They sharpen Project 2 work in flight, and they seed next steps (and next-cohort topics). They are never examined.

**Why the presentation comes after the draft.** Presenting design *plus preliminary results* means every talk has substance, and the feedback — the questions and discussion in the room — arrives exactly one revision week before the final deadline, when teams can still act on it.

**Not comfortable presenting your own idea?** That is allowed for. If you would rather not share your novel idea publicly before submission, you may instead present **one closely related paper, end to end** — its problem, method, evidence, and limitations, plus one slide on how it connects to your project's area. Same slot, same length, same rubric; tell the instructor by the Fri Dec 4 draft deadline so the sessions can be ordered.

**Where the slack is — and is not**

The semester has **40 class meetings** (Fri Oct 30 is a deadline day, not a class): 23 in Module 1's weeks, 8 in Module 2's — two of them online — and 9 in the Weeks 12–14 finish block. The advanced-topic sessions (Nov 20 – Dec 2) are the slack: each one yields to a Project 2 clinic, an overflow presentation slot, or a snow-day catch-up before anything else moves. Everything before Week 12 is committed.

Presentations run five talks per 50-minute session (8 min + 2 questions), so Dec 7 and Dec 9 hold ten. With solo-or-pair teams a class of twenty can produce up to fifteen talks: talks 11–15 go to **Wed Dec 2**, whose advanced-topic slot converts to a third presentation session, and only past fifteen does the Fri Dec 4 clinic yield. The final Friday stays free for the wrap and the deadline.

A voluntary demo hour in finals week (Dec 14–18) remains available — this course has no final exam.

Each topic has more material prepared than its sessions allow; the slides are built to be selected from, not marched through.

**Grading**

| Component                                                | Weight     |
|----------------------------------------------------------|------------|
| Midterm exam 1 (Module 1)                                | 10%        |
| Midterm exam 2 (Module 2)                                | 10%        |
| Participation & in-class engagement                      | 10%        |
| Project 1 — Build it, use it, break it, assure it        | 25%        |
| Project 2 — Research project (conference submission)     | 45%        |
| <b>Bonus — workshop submission</b> (instructor-approved) | <b>+5%</b> |

**There is no final exam.** The two midterms are short, in-class exams: Midterm 1 covers the Module 1 foundations lectures taught through the Week 6 review (through reasoning — the Weeks 7–8 material on retrieval, agents and evaluation is *not* on it); Midterm 2 covers the Module 2 lectures. The ⚙️ AI-assisted/agentive SE sessions are practice-oriented and are assessed through the projects, never on an exam. Each examinable lecture ends with a 'Key takeaways' slide; those slides are the examinable core.

**Bonus credit (+5% of the course grade).** A team whose final paper is **reviewed and approved by the instructor** submits it to a suitable **conference workshop or similar venue** (e.g., an ICSE/FSE workshop,

Alware, or an MSR/EASE track — chosen with the instructor over the winter break; the realistic deadlines fall January–March) and earns bonus credit for every team member. **The target is deliberately a workshop, not a top-tier main track** — a workshop paper of up to **8 pages (not including references)** is what one semester honestly produces, and a good workshop with engaged reviewers teaches more than a main-track rejection. The bonus is awarded for a completed, instructor-approved submission — **not** for acceptance, which is decided long after grades are due. Approval means the paper meets a real venue's bar: a falsifiable research question, a fair baseline, reported variance, an honest threats-to-validity section, and compliance with the venue's formatting and AI-disclosure rules. Teams that are not approved lose nothing; the bonus only ever adds.

**Participation & in-class engagement (10%)**

- **Sign in, and you have it.** Attendance is the whole requirement — sign the sheet each class. There is no discussion quota and nobody keeps score of how often you speak.
- Discussion is **assumed, not graded**: the paper discussions, the hands-on sessions, the pooled-findings session and the design-presentation Q&A are how this class runs, and everyone is expected to join in naturally.
- **One way to lose points: disrespect.** Every student's questions, answers and presentations are treated with respect. Dismissing or belittling another student's contribution costs participation credit.

**Project 1 — Build It, Use It, Break It, Assure It (25%, individual, Weeks 1–9)**

Full spec: the [Project 1 page](#) on the course website

**The question Project 1 answers:** *how should software engineers be trained when AI is part of the process?* The course's answer is that you make them build the component, use it for real work, attack it, and then engineer the checks that catch what they found.

**One deadline: everything is due Fri, Oct 30** — one bundle: model card + repo, task report, findings table, QA report + a **3-minute recorded demo video**, AI-use log + reflection. The four phases have a **recommended pace**, not graded deadlines:

| Phase      | Recommended pace | What you do                                                                                                     | Hand in (Oct 30)                   |
|------------|------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1 · Build  | by Fri Sep 25    | Train a small code model with nanoGPT, then <b>one ablation</b> — remove a part, predict the effect, measure it | Model card (incl. ablation) + repo |
| 2 · Use    | by Fri Oct 9     | Use <i>your own</i> model on a real SE task, on 20 real inputs                                                  | Task report                        |
| 3 · Break  | by Fri Oct 16    | Red-team it — find 10+ distinct failures                                                                        | <b>Findings table</b>              |
| 4 · Assure | by Fri Oct 23    | Build checks that catch them; measure your catch rate                                                           | QA report + demo video             |

The pace exists because the later phases cannot be compressed into the last weekend — and because the **Week 8 pooled-findings session (Fri, Oct 23)** works from whatever findings table you bring that day.



Everyone uses the same eight failure categories, so the class ends the term with a pooled catalogue of how a code model fails and how much of that ordinary testing catches; the final catalogue is assembled from the Oct 30 submissions.

The AI-assisted workflow is **required**, with a log and a one-page reflection.

**Graded on what you found and what you built to catch it — not on model quality.** A model that produces broken Python is fine; it gives you more to find in Phase 3. Rubric: the [Project 1 rubric](#).

**A note on data.** Anonymised Phase 3 findings may be used in a study of how this course is taught. Participation is voluntary, opt-in via a Week 1 consent form, and has no bearing on grades in either direction.

**Compute:** the free GPU tiers (Kaggle, Colab and similar) are the working plan and are enough for this model; an ACCESS classroom allocation has been applied for and will be announced if granted — see the [compute guide](#).

## **Project 2 — Research Project with Conference Submission (45%, solo or in pairs, Weeks 3–14)**

Full spec: the [Project 2 page](#) on the course website, with the [proposal template](#), the suggested [reading list](#), and the candidate venues.

An original research project on software engineering with/for generative AI, carried out **solo or in pairs** and written up as a conference-format paper. The goal is to take one question from idea to evidence to a paper an outside reviewer would take seriously — and then actually submit it. The rubric is the same for one student as for two; scope expectations scale with the number of hands, and the proposal is where scope is agreed. Undergraduates are encouraged to pair with a graduate student (see above — a suggestion, not a requirement).

**The research thread starts in Week 3.** Topics circulate on **Fri, Sep 18**, and by **Fri, Oct 2** each student files a half-page **research radar**: solo-or-pair declared, two candidate directions, and three papers actually opened, one sentence each. The point is to put you in research mode early — reading the literature in September, while Project 1 is teaching you the craft. Your Phase 3 findings table will later sharpen the choice (revisit your radar after red-teaming — several of the best questions come from there). Full-time execution begins Nov 2, giving **five working weeks** before the draft. Many projects extend the Project 1 experience directly — four of the circulated candidate topics reuse the class's own models, probes, and QA checks.

**Why we submit to a real venue.** Writing for an external audience is the pedagogy, not a trophy hunt. A real deadline and real reviewers force the things this course is about: a question you can actually answer, an honest baseline, results you can defend, and limitations you state yourself. Teams that make it that far learn more than any term paper could teach.

**How submission works — and what the target is.** The target venue is a **conference workshop** (ICSE/FSE workshops, Alware, MSR/EASE tracks and similar), on purpose: a workshop paper of up to 8 pages (not including references) is the right-sized ambition for one semester, and setting that expectation up front is healthier than chasing a top-tier main track. Papers are **submission-ready on Dec 11**; teams may post an arXiv preprint then. Over the winter break the instructor works with each team to pick the workshop whose scope and deadline fit their work — options are discussed in Week 12, and the deadlines

that suit a December finish fall between January and March. **Papers the instructor reviews and approves are submitted for +5% bonus credit** (see Grading).

**Grading is independent of the outcome.** Your grade reflects the quality of the research and the writing — the study you designed, the evidence you gathered, the honesty of your analysis. Acceptance decisions arrive months after grades are final and play no part in them. A rigorous negative result earns full marks.

Milestones (weights are of the Project 2 grade):

| Date          | Milestone                                                                                              | Weight           |
|---------------|--------------------------------------------------------------------------------------------------------|------------------|
| Fri, Sep 18   | Topics circulated (posted on the course website) — the research thread opens                           | —                |
| Fri, Oct 2    | <b>Research radar</b> — half a page: solo-or-pair, two candidate directions, three papers skimmed      | 5                |
| Mon, Oct 26   | Five-minute topic pitch, in class                                                                      | 5                |
| Fri, Oct 30   | Written proposal (problem, method, baseline, evaluation plan) — topic pivots allowed until Fri, Nov 20 | 15               |
| Nov 2 – Dec 4 | Execution — five working weeks; the ⚙ advanced sessions run alongside; check-ins in office hours       | —                |
| Fri, Dec 4    | Full paper draft                                                                                       | 15               |
| Dec 7 / Dec 9 | Design presentation + preliminary results                                                              | 15               |
| Fri, Dec 11   | Final paper + artifact, submission-ready                                                               | 45               |
| after         | Instructor review over break → workshop chosen with each team; approved papers submitted               | +5% course bonus |

## Course Policies

### Use of AI tools

Using AI assistants (Copilot, Claude, ChatGPT, agents, etc.) is **required for Project 1 and encouraged everywhere else** — disciplined use of these tools is a learning outcome of this course. Requirements:

- **Disclose** AI use in every deliverable (which tools, for what, and how outputs were validated); Project 1 additionally requires an AI-use log.
- **You are accountable** for everything you submit: AI-generated code and text must be reviewed, tested, and understood. "The AI wrote it" is not a defense for plagiarized, incorrect, or insecure work.
- AI tools are **not permitted during the two midterm exams**.
- For Project 2, follow the target venue's policy on AI-generated text in submissions.

**Late work**

Project milestones lose 10% per day late (max 3 days) unless arranged in advance. The Dec 4 draft and Dec 11 final deadlines are firm because the presentation sessions and the submission timeline depend on them.

**Academic integrity**

All work is governed by the MTU Academic Integrity Policy. Collaboration within teams is expected; representing others' (or undisclosed AI) work as your own analysis is not.

**Accessibility and wellbeing**

MTU complies with the ADA and Section 504. Students with disabilities should contact Student Accessibility Services and the instructor early in the semester. Standard MTU syllabus statements (accessibility, non-discrimination, Title IX, mental health resources) apply and will be linked on Canvas.

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